THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The angle of incidence is always equal to the angle of reflection. This is the 1

(a) law of refraction **(b)** second law of reflection
(c) Snell's law **(d)** first law of reflection

2. The image in the human eye is formed on the 1

(a) iris **(b)** cornea **(c)** retina **(d)** pupil

3. The SI unit of electric charge is 1

(a) coulomb **(b)** volt **(c)** ohm **(d)** ampere

4. The magnetic field lines outside a bar magnet are directed 1

(a) along the magnet only **(b)** in no definite direction
(c) from the north pole to the south pole **(d)** from the south pole to the north pole

5. The refractive index of a medium is the ratio of the speed of light 1

(a) in air to that in water only **(b)** in glass to that in water only
(c) in vacuum to that in the medium **(d)** in the medium to that in vacuum

6. Myopia is corrected by using 1

(a) a plane mirror **(b)** a convex lens
(c) a concave lens **(d)** a cylindrical lens

7. Ohm's law states that the current through a conductor is 1

(a) independent of the potential difference
(b) inversely proportional to the potential difference
(c) directly proportional to the potential difference, at constant temperature
(d) proportional to the square of V

8. The field pattern of a solenoid is similar to that of 1

(a) a bar magnet **(b)** a plane mirror
(c) a horseshoe magnet only **(d)** a straight wire

9. A convex lens is also called 1

(a) a diverging lens **(b)** a cylindrical lens
(c) a plane lens **(d)** a converging lens

10. Splitting of white light into its seven colours is called 1

(a) reflection **(b)** dispersion **(c)** refraction **(d)** scattering

11. For resistors in series, the equivalent resistance is 1

(a) R1 + R2 + R3 **(b)** less than each

(c) the average of the resistances **(d)** 1/R1 + 1/R2

12. Electromagnetic induction is the production of 1

(a) force on a coil

(b) a magnetic field by a steady current

(c) heat by current

(d) induced current in a coil due to a changing magnetic field

13. The image formed by a plane mirror is 1

(a) real and diminished

(b) real and inverted

(c) virtual, erect and of the same size as the object

(d) virtual and magnified

14. Most of the refraction of light entering the eye takes place at the 1

(a) iris **(b)** retina

(c) outer surface of the cornea **(d)** eye lens

15. The SI unit of electric current is 1

(a) coulomb **(b)** ampere **(c)** watt **(d)** volt

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. State the laws of reflection of light. 2

17. What is meant by the power of accommodation of the eye? 2

18. Define one ampere of current. 2

19. State the rule to find the direction of the magnetic field around a straight current-carrying wire. 2

20. State the sign convention for the focal length of concave and convex mirrors. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. List the characteristics of the image formed by a plane mirror. 3

22. Describe the structure and function of the cornea, iris, pupil and retina. 3

23. A current of 0.5 A flows in a conductor for 10 minutes. Find the charge that flows and the number of electrons. 3

24. Draw (describe) the magnetic field lines around a bar magnet and state three properties. 3

OR

A convex lens has focal length 10 cm. An object is placed 30 cm from it. Find the image distance and magnification. 3

25. Why is it difficult for older persons to see nearby objects? 3

OR

Find the equivalent resistance when 2 ohm, 3 ohm and 5 ohm are connected in series, and the current from a 6 V battery. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the images formed by a concave mirror for different positions of the object (at infinity, beyond C, at C, between C and F, at F, between F and P) with nature and size. 5

OR

Draw the structure of the human eye (describe) and explain the function of its parts: cornea, iris, pupil, lens, retina, optic nerve. 5

27. State and explain Ohm's law. Describe an activity to verify it and how the V-I graph is used to find the resistance. 5

OR

Describe an activity to show that a current-carrying wire produces a magnetic field around it. State the right-hand thumb rule. 5

END OF QUESTION PAPER Total: 50 Marks

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